

TEST CASES DOCUMENT

Project: AI-Powered Schema Aware Test Data Generator

TC-001: Load Valid SQL Schema

Module: Schema Input

Objective: Verify that a valid SQL schema is loaded successfully.

Input:

```
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100)  
);
```

Expected Result:

- Schema accepted successfully
- Table count detected as 1

Actual Result:

- Schema loaded successfully

Status: PASS

TC-002: Parse Multi-Table Schema

Module: Schema Parser

Objective: Verify detection of multiple tables and relationships.

Expected Result:

- Tables detected correctly
- Foreign key relationship identified

Actual Result:

- 2 tables detected
- Relationship identified successfully

Status: PASS

TC-003: Generate Synthetic Data

Module: Data Generator

Input:

- Customers = 100 Records

Expected Result:

- 100 records generated successfully

Actual Result:

- 100 records generated

Status: PASS

TC-004: CSV Export

Module: Export Engine

Objective: Verify CSV export functionality.

Expected Result:

- CSV file downloaded successfully

Actual Result:

- CSV generated and downloaded correctly

Status: PASS

TC-005: End-to-End Workflow

Module: System Integration

Workflow:

1. Load Schema
2. Analyze Schema
3. Generate Data
4. Validate Data
5. Export CSV

Expected Result:

- Workflow completes without errors

Actual Result:

- Workflow executed successfully

Status: PASS

TC-006: Invalid SQL Schema Handling

Module: Schema Validation

Input:

CREATE TABLE users (
id INT PRIMARY KEY

Expected Result:

- Error message displayed

Actual Result:

- Application failed to show detailed error message

Status: FAIL

TC-007: Large Dataset Generation

Module: Data Generator

Input:

- 100000 Records

Expected Result:

- Dataset generated successfully

Actual Result:

- Generation process timed out

Status: FAIL

TC-008: Foreign Key Validation

Module: Relationship Manager

Objective: Verify referential integrity validation.

Expected Result:

- Invalid foreign keys detected

Actual Result:

- Some orphan records were not detected

Status: FAIL

TC-009: SQL Export Generation

Module: Export Engine

Objective: Verify SQL INSERT statement generation.

Expected Result:

- SQL file generated correctly

Actual Result:

- Export failed for schemas with multiple dependencies

Status: FAIL

TC-010: Empty Schema Submission

Module: Input Validation

Input:

- Empty schema input

Expected Result:

- Warning message displayed

Actual Result:

- Application attempted processing instead of validation

Status: FAIL

Test Summary

Status Count

PASS 5

FAIL 5

TOTAL 10

Overall Result

- Successful Test Cases: 5
- Failed Test Cases: 5
- Pass Percentage: 50%

The application successfully performs core schema parsing, data generation, validation, and export functionalities. However, improvements are required in error handling, scalability, foreign key validation, and export reliability for complex schemas.

Data generation completed.

Validation Results

Validation Passed

Issues Found

Tables Generated

Yes

0

3

Data Preview

books members loans

Rows: 50 | Columns: 4

	book_id		title	author	isbn
2		3	Malgudi Days	Priya Reddy	978-81-1002-102-0
3		4	The White Tiger	Arjun Kumar	978-81-1003-103-0
4		5	The Guide	Ananya Iyer	978-81-1004-104-0
5		6	Train to Pakistan	Karthik Nair	978-81-1005-105-0
6		7	Malgudi Days	Sneha Patel	978-81-1006-106-0
7		8	The White Tiger	Rohit Gupta	978-81-1007-107-0

Deploy

Validation Results

Validation Passed

Issues Found

Tables Generated

Yes

0

2

Data Preview

users teams

Rows: 50 | Columns: 3

	user_id		name	team_id
0		1	Varalakshmi TC	1
1		2	Rahul Sharma	2
2		3	Priya Reddy	3
3		4	Arjun Kumar	4
4		5	Ananya Iyer	5
5		6	Karthik Nair	6
6		7	Sneha Patel	7
7		8	Rohit Gupta	8
8		9	Divya Rao	9
9		10	Vikram Menon	10